Types of Neural Networks in Deep Learning

Deep Learning mainly uses different types of **Neural Networks** for different tasks.

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1. Artificial Neural Network (ANN)

Basic neural network.

Used for:

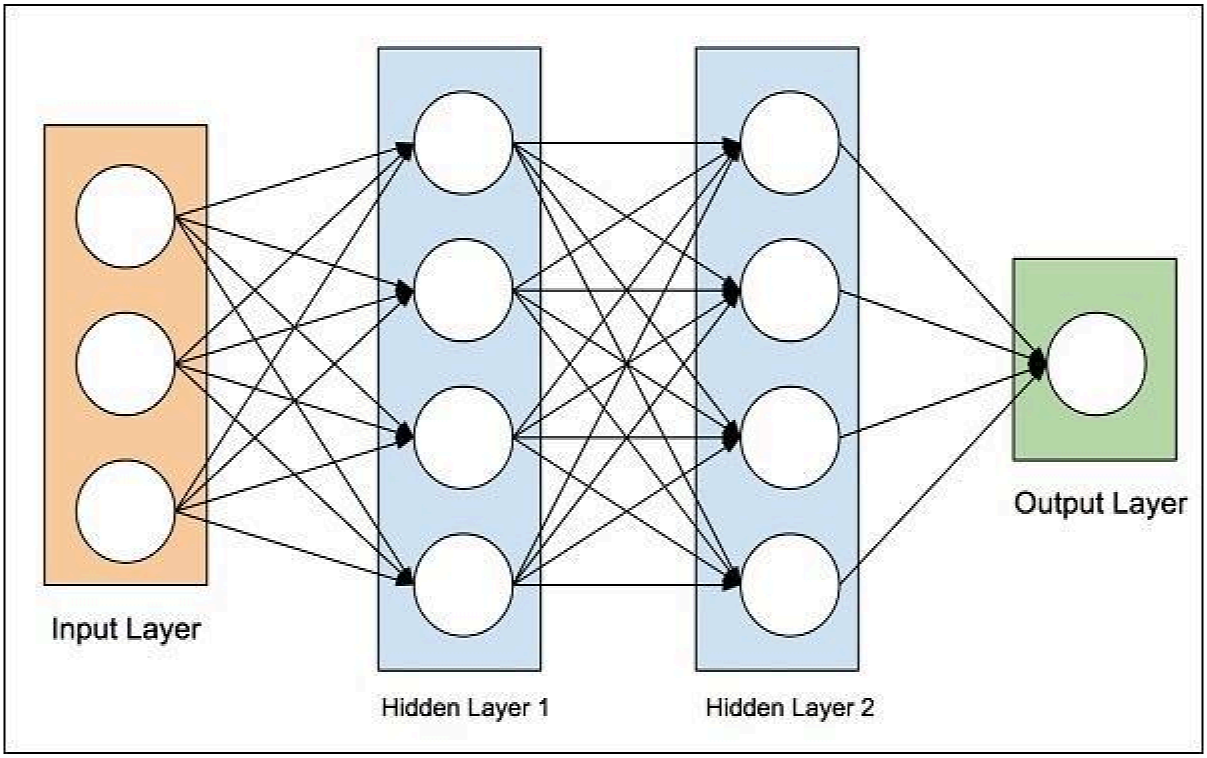
- Prediction
- Classification

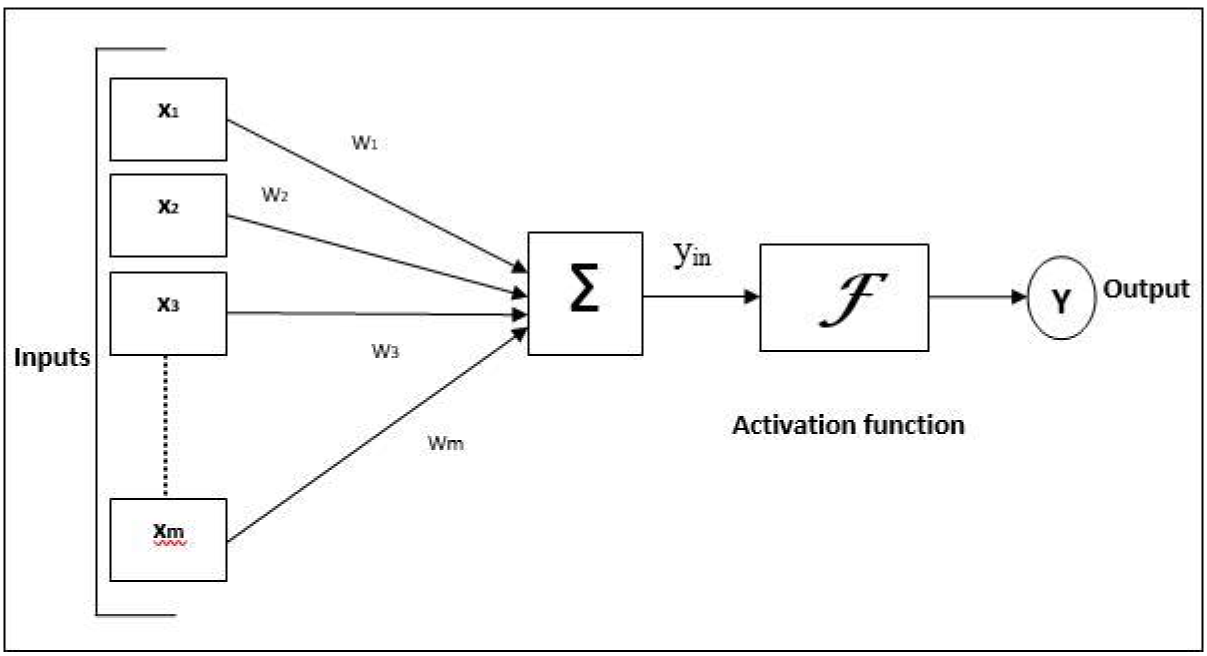
Example:

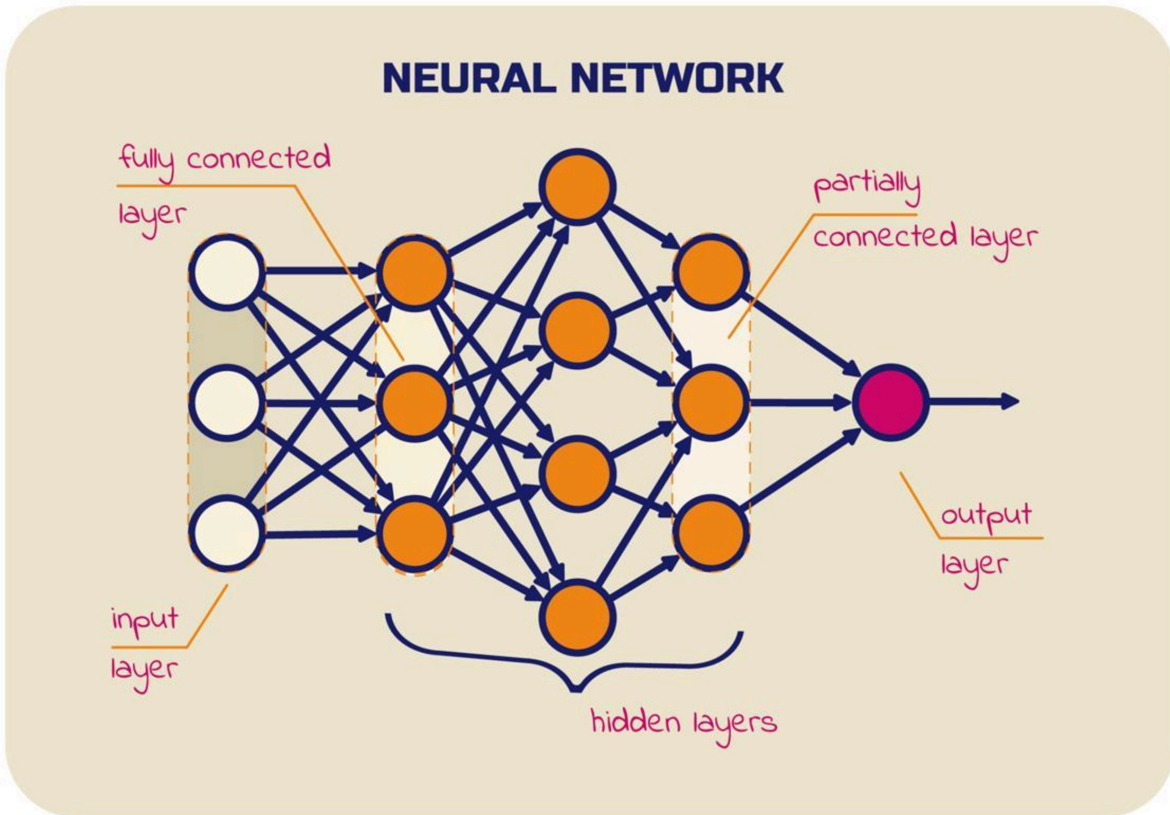
Student pass/fail prediction

Structure:

Input Layer → Hidden Layer → Output Layer


Input Layer
Hidden Layer 1
Hidden Layer 2
Output Layer


Inputs
x1
x2
x3
xm
w1
w2
w3
wm
Σ
yin
Activation function
Y
Output



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2. Convolutional Neural Network (CNN)

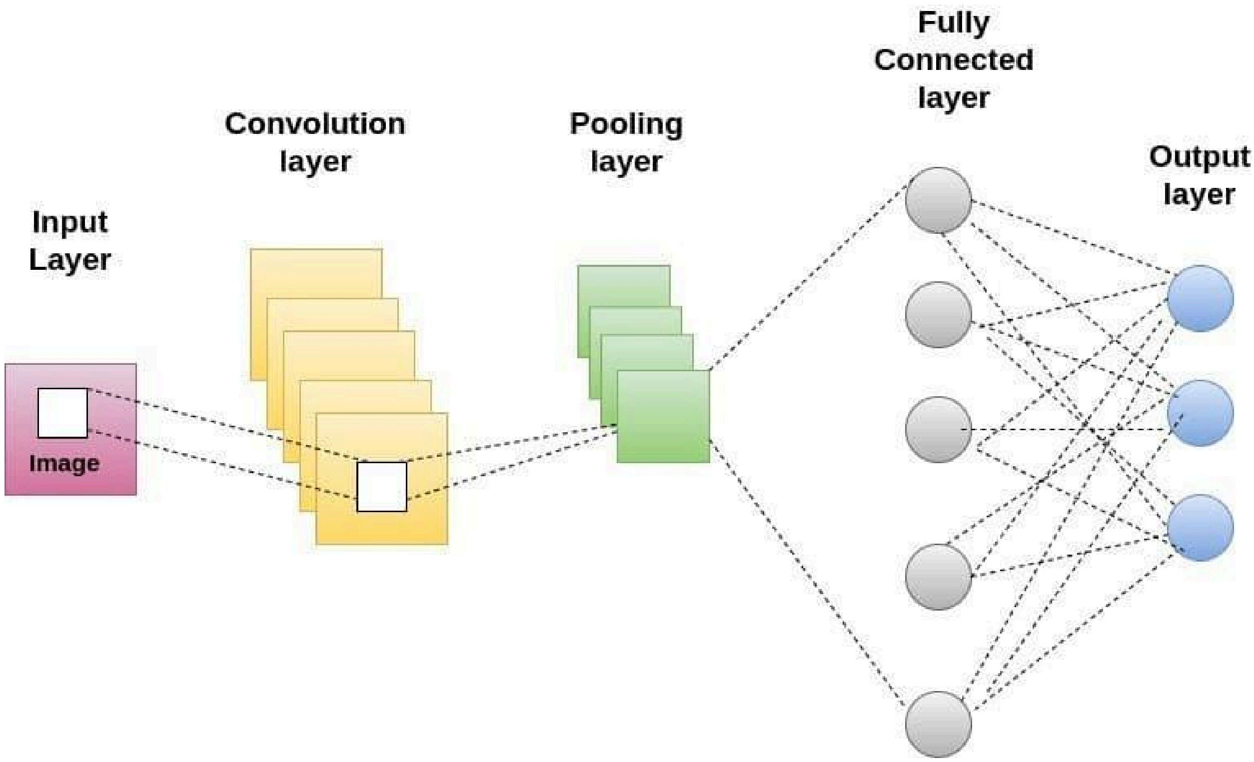
Mainly used for images and videos.

Used for:

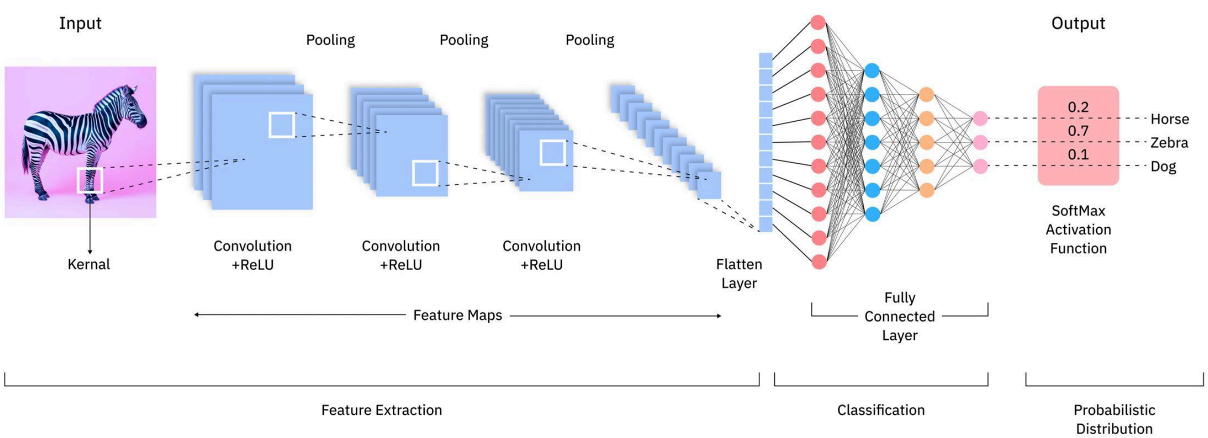
- Face recognition
- Object detection
- Medical image analysis

Example:

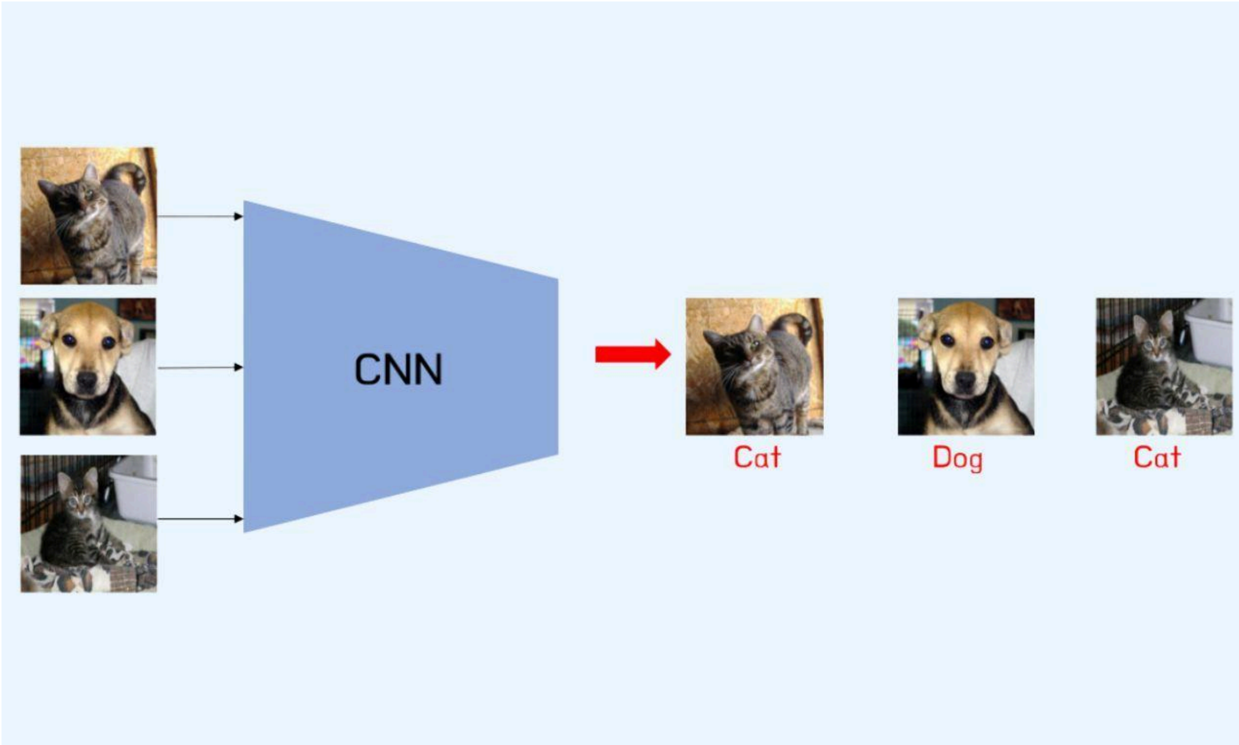
Detect cat or dog from image



Input Layer
Image
Convolution layer
Pooling layer
Fully Connected layer
Output layer



Input
Pooling
Pooling
Pooling
Output
0.2
0.7
0.1
Horse
Zebra
Dog
SoftMax Activation Function
Kernal
Convolution +ReLU
Convolution +ReLU
Convolution +ReLU
Flatten Layer
Fully Connected Layer
Feature Maps
Feature Extraction
Classification
Probabilistic Distribution



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3. Recurrent Neural Network (RNN)

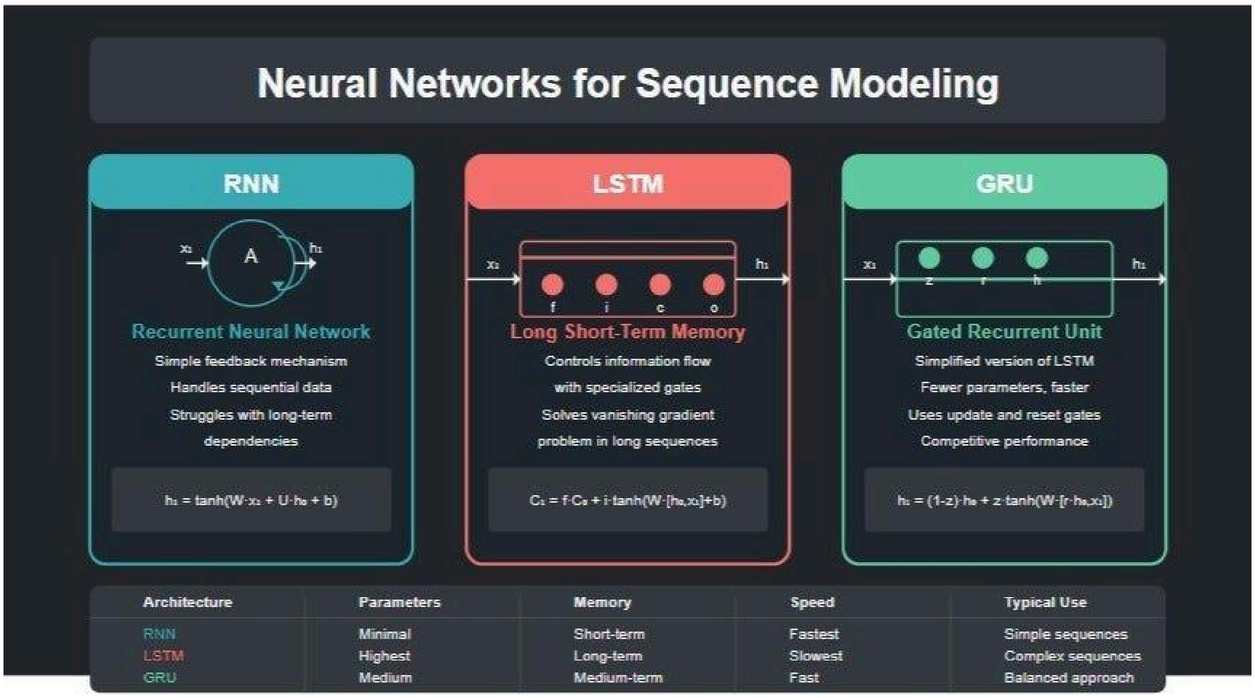
Used for sequential data.

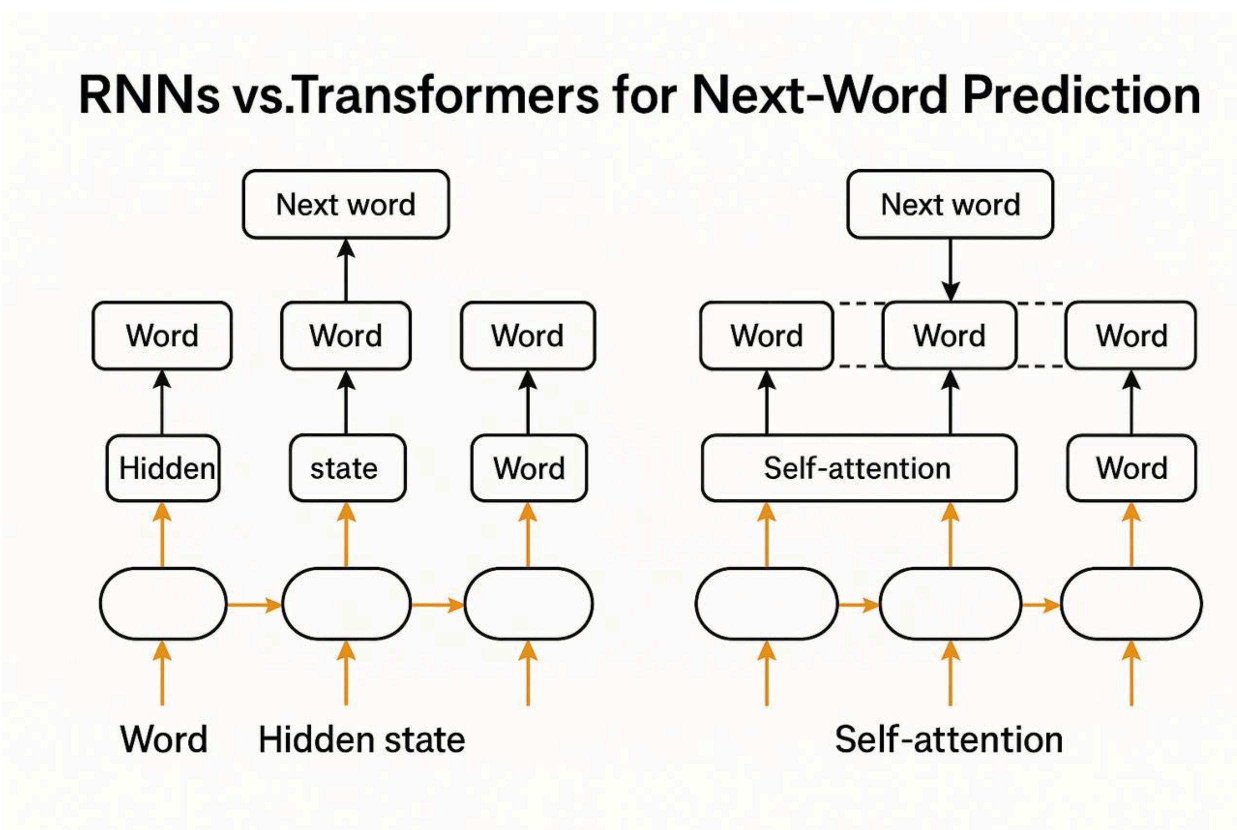
Used for:

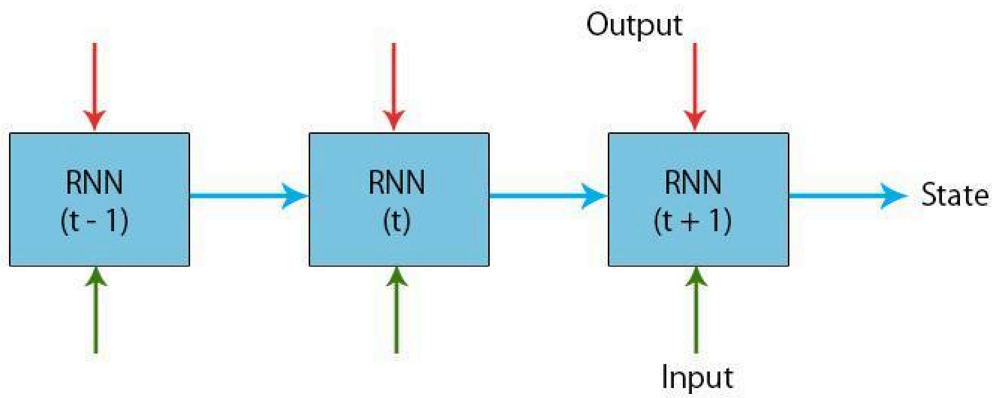
- Speech recognition
- Language translation
- Text prediction

Example:

Predict next word while typing


Neural Networks for Sequence Modeling
RNN
A
Recurrent Neural Network
Simple feedback mechanism
Handles sequential data
Struggles with long-term
dependencies
$h_t = \tanh(W \cdot x_t + U \cdot h_{t-1} + b)$
LSTM
f i c o
Long Short-Term Memory
Controls information flow
with specialized gates
Solves vanishing gradient
problem in long sequences
$C_t = f \cdot C_{t-1} + i \cdot \tanh(W \cdot [h_{t-1}, x_t] + b)$
GRU
z r h
Gated Recurrent Unit
Simplified version of LSTM
Fewer parameters, faster
Uses update and reset gates
Competitive performance
$h_t = (1-z) \cdot h_{t-1} + z \cdot \tanh(W \cdot [r \cdot h_{t-1}, x_t])$
Architecture | Parameters | Memory | Speed | Typical Use
RNN | Minimal | Short-term | Fastest | Simple sequences
LSTM | Highest | Long-term | Slowest | Complex sequences
GRU | Medium | Medium-term | Fast | Balanced approach


RNNs vs.Transformers for Next-Word Prediction
Next word
Word
Word
Word
Hidden
state
Word
Word
Hidden state
Next word
Word
Word
Word
Self-attention
Word
Self-attention



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4. Long Short-Term Memory (LSTM)

Special type of RNN.

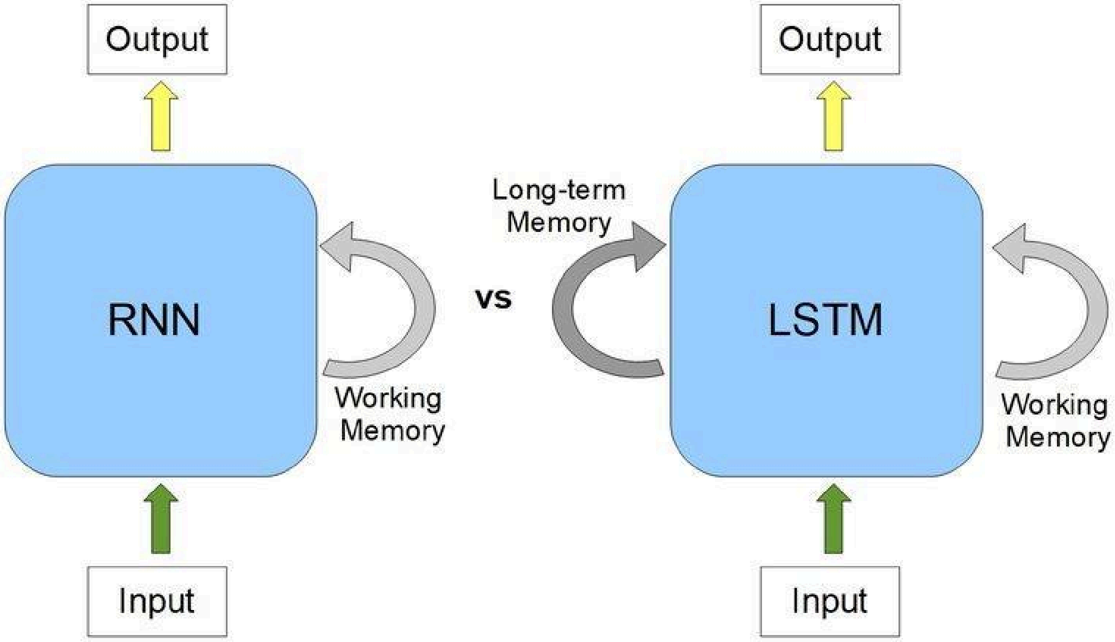
Better at remembering long information.

Used for:

- Chatbots
- Stock prediction
- Weather forecasting

Example:

Remember previous words in a sentence


Output
Output
RNN
vs
Long-term
Memory
LSTM
Working
Memory
Working
Memory
Input
Input

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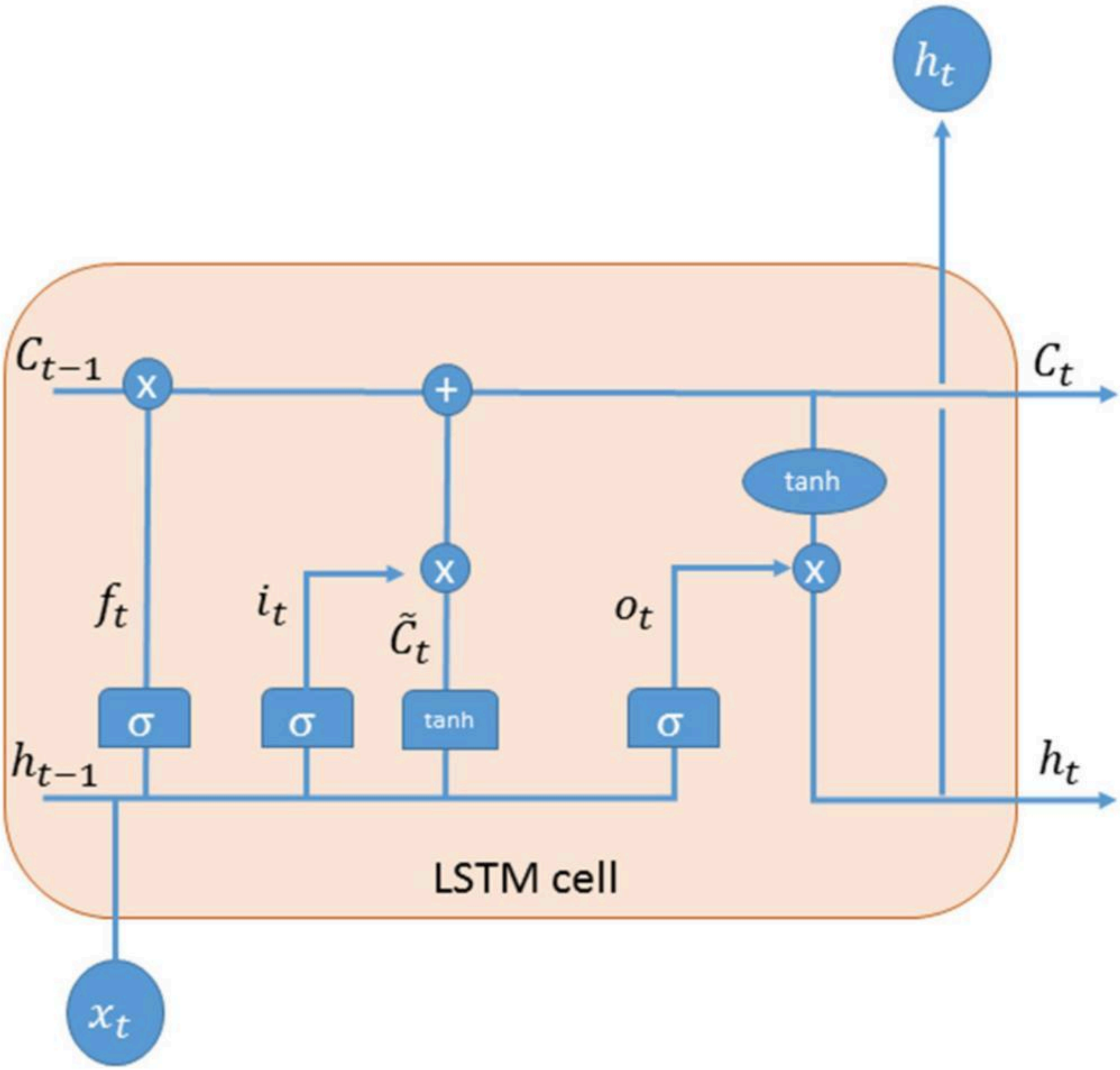
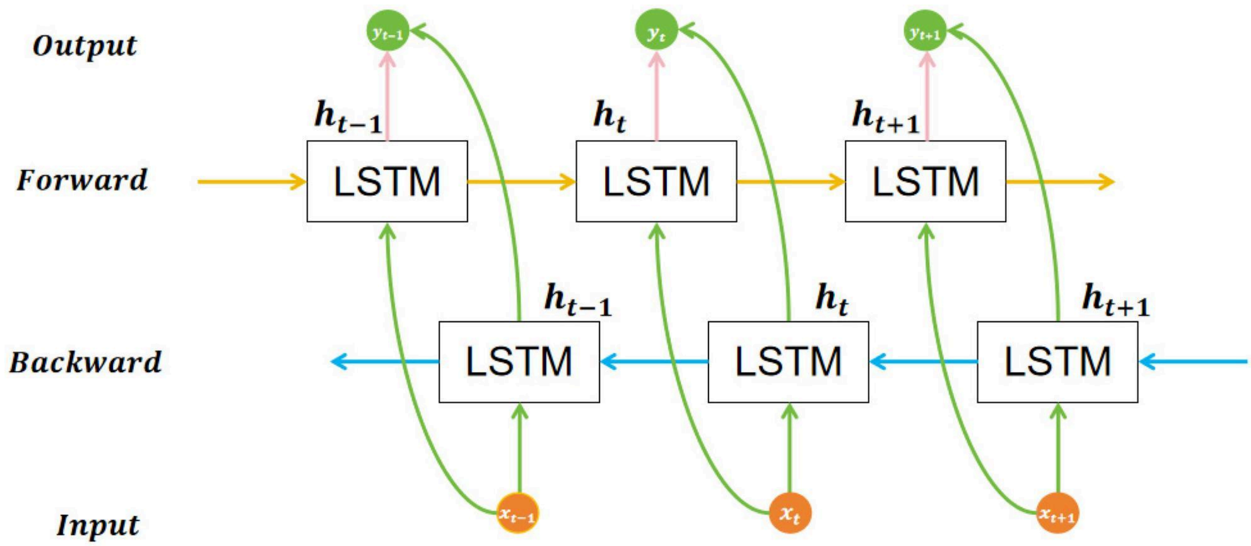


h_t
C_{t-1}
x
+
C_t
tanh
x
x
f_t
i_t
$\tilde{C}_t$
o_t
σ
σ
tanh
σ
h_{t-1}
h_t
LSTM cell
x_t



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5. Generative Adversarial Network (GAN)

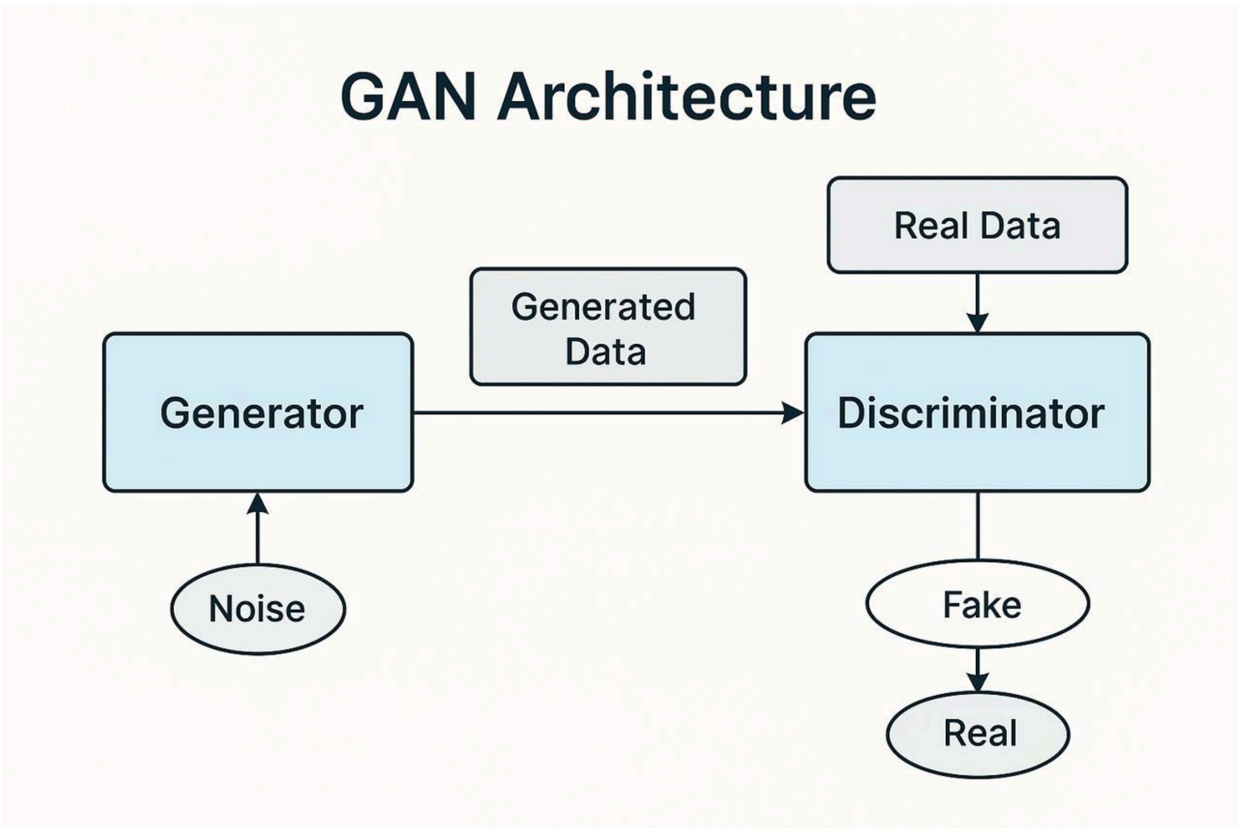
Two networks compete with each other.

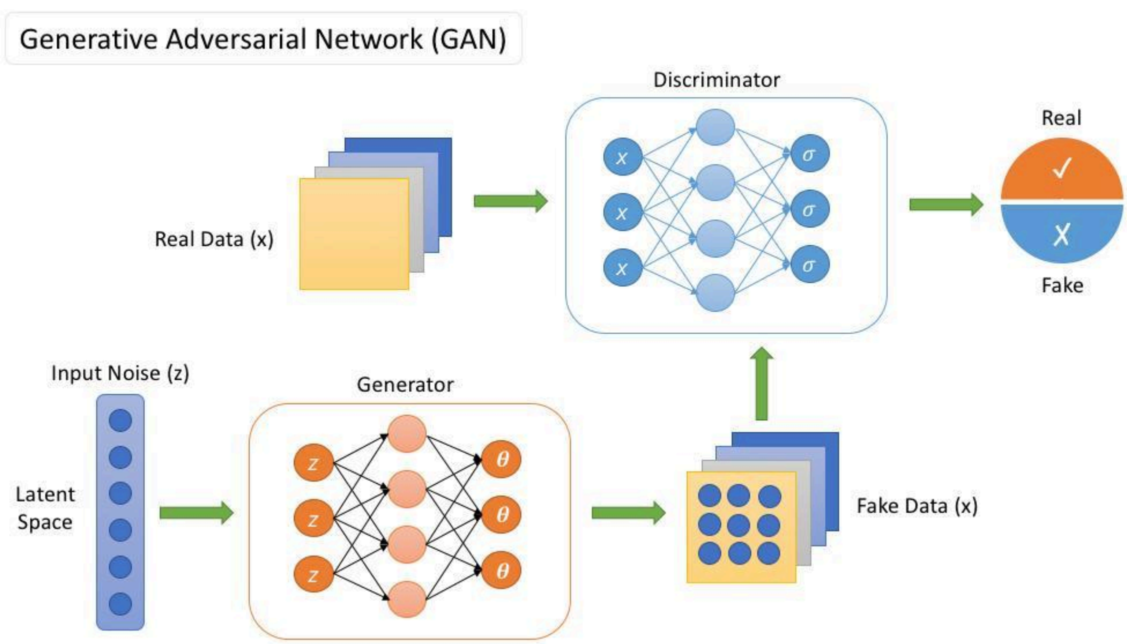
Used for:

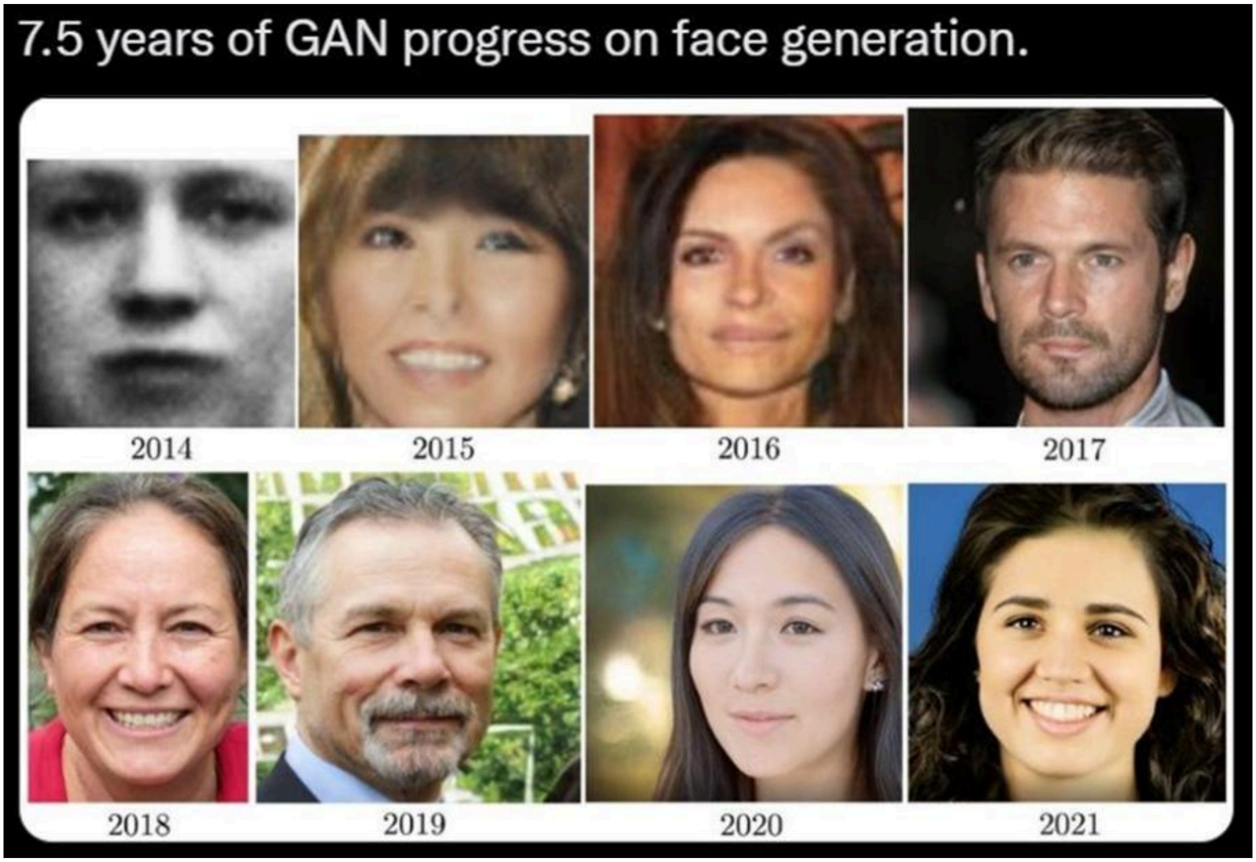
- AI image generation
- Deepfake
- Photo enhancement

Example:

Generate human face images


GAN Architecture
Real Data
Generated
Data
Generator
Discriminator
Noise
Fake
Real


Generative Adversarial Network (GAN)
Discriminator
Real Data (x)
Real
Fake
Input Noise (z)
Latent
Space
Generator
Fake Data (x)



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6. Transformer Network

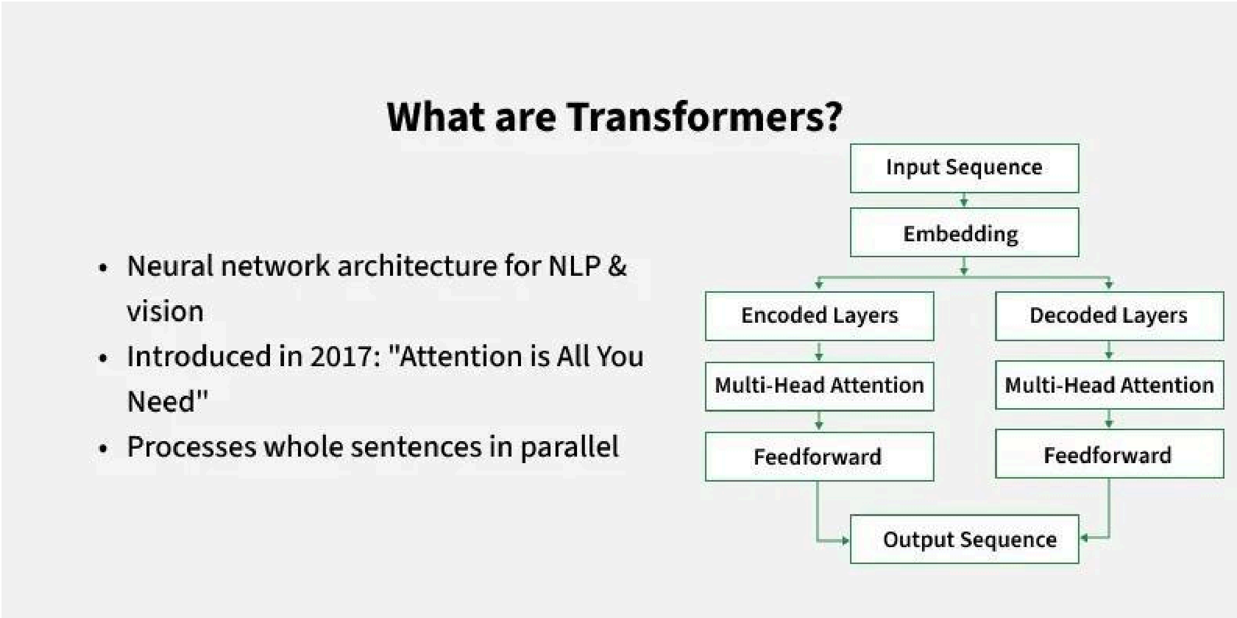
Modern powerful neural network.

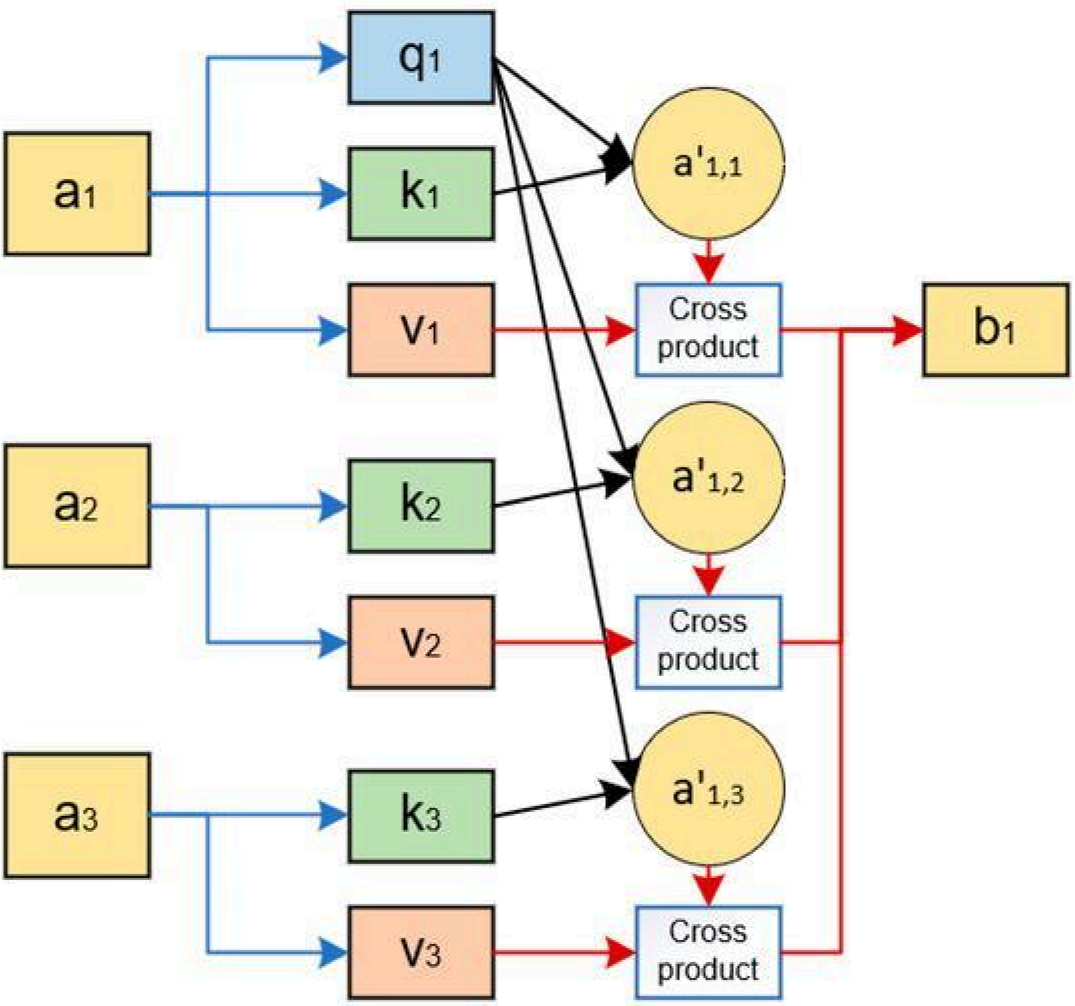
Used for:

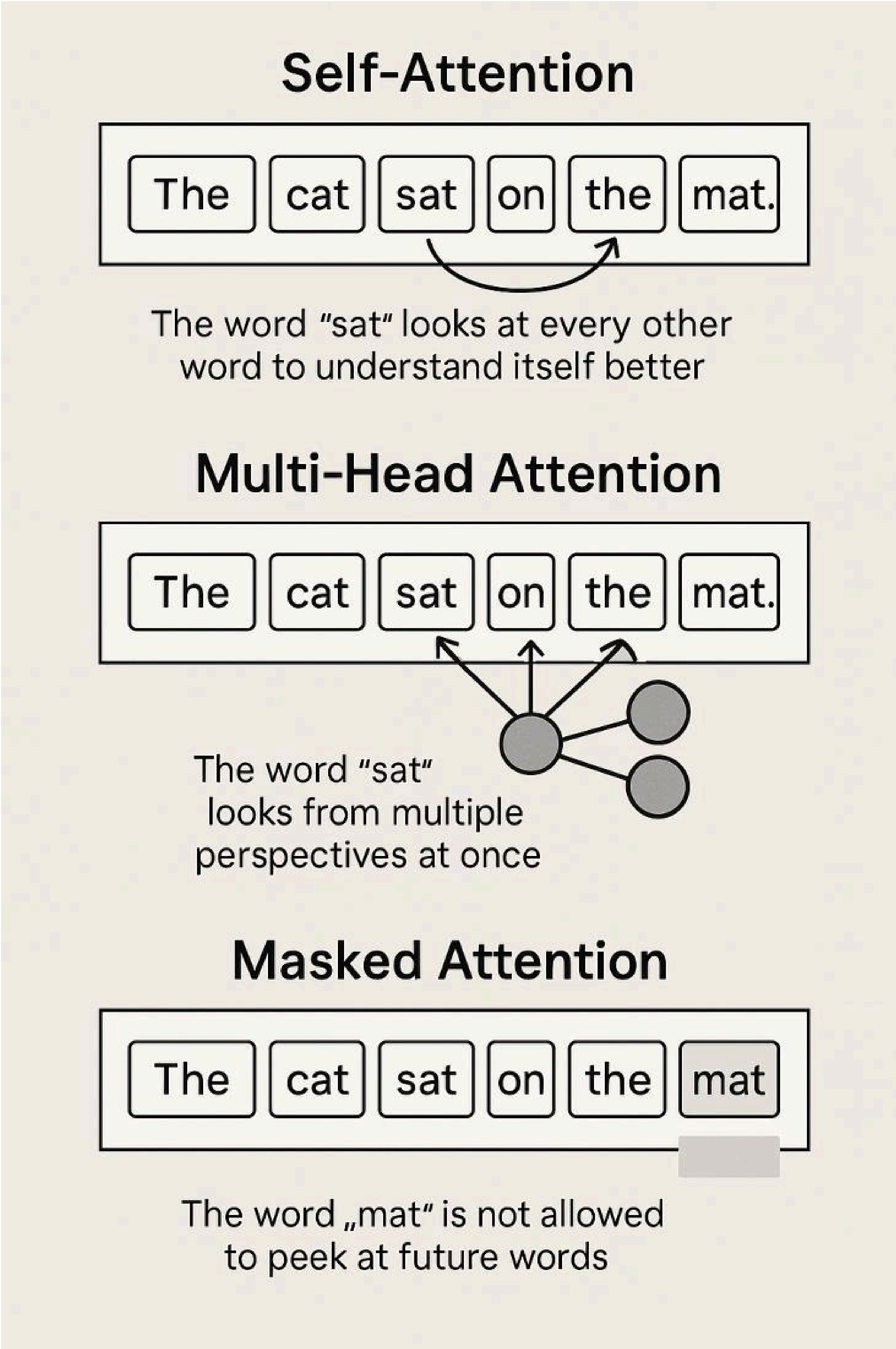
- Chatbots
- Translation
- AI assistants

Example:

Used in OpenAI models like ChatGPT


What are Transformers?
Neural network architecture for NLP & vision
Introduced in 2017: "Attention is All You Need"
Processes whole sentences in parallel
Input Sequence
Embedding
Encoded Layers
Decoded Layers
Multi-Head Attention
Multi-Head Attention
Feedforward
Feedforward
Output Sequence


q1
a1
k1
v1
a'1,1
Cross product
b1
a2
k2
v2
a'1,2
Cross product
a3
k3
v3
a'1,3
Cross product


Self-Attention
The
cat
sat
on
the
mat.
The word "sat" looks at every other word to understand itself better
Multi-Head Attention
The
cat
sat
on
the
mat.
The word "sat" looks from multiple perspectives at once
Masked Attention
The
cat
sat
on
the
mat
The word „mat" is not allowed to peek at future words

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Simple Summary Table

Neural Network	Mainly Used For
ANN	Basic prediction
CNN	Images
RNN	Sequential/Text data
LSTM	Long memory sequences
GAN	Image generation
Transformer	Modern AI/chatbots

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Easy Memory Trick

ANN → Basic
CNN → Camera/Image
RNN → Reading sequence
LSTM → Long memory
GAN → Generate new data
Transformer → Talking AI

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Difference Between Neural Networks in Deep Learning

Neural Network	Full Form	Main Purpose	Best For	Example	Special Feature
ANN	Artificial Neural Network	Basic learning and prediction	Simple structured data	Student pass/fail prediction	Simple network with input, hidden, output layers
CNN	Convolutional Neural Network	Image processing	Images & videos	Cat vs dog detection	Automatically detects image features
RNN	Recurrent Neural Network	Sequential data processing	Text, speech, time series	Next word prediction	Remembers previous inputs
LSTM	Long Short-Term Memory	Long sequence memory	Long text/speech sequences	Chatbot conversations	Avoids forgetting old information
GAN	Generative Adversarial Network	Generate new data	AI image/video generation	Fake human face generation	Two networks compete (Generator vs Discriminator)
Transformer	Transformer Network	Understanding relationships in data	NLP, chatbots, translation	ChatGPT	Uses attention mechanism for better understanding

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Simple Real-Life Comparison

Neural Network	Real-Life Analogy
ANN	Student solving basic math problems

CNN	Human eyes recognizing objects
RNN	Reading sentence word by word
LSTM	Remembering long story details
GAN	Artist creating realistic fake paintings
Transformer	Smart person understanding full conversation context

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Easy Shortcut

ANN → Basic prediction
CNN → Images
RNN → Sequence
LSTM → Long memory
GAN → Generate content
Transformer → Smart AI language understanding